

PinPilot Wiring Documentation

Smart Farm Node

Board: ESP32 DevKit (WROOM-32)

Generated by **PinPilot** (IBM Bob 2.0 Hackathon)

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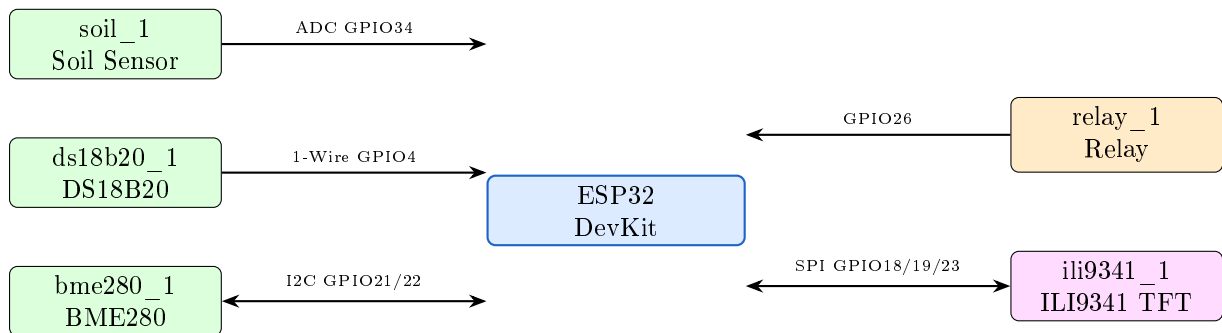
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1 Project Summary

This document describes the wiring, pin assignment, bus configuration, and LVGL dashboard layout for the **Smart Farm Node** demo project. The target board is the **ESP32 DevKit (WROOM-32)**.

Property	Value
Board	ESP32 DevKit (WROOM-32)
Components	5
Buses	i2c0, spi0
WiFi enabled	Yes
Serial logging	Yes
LVGL version	9.2.2
RAM usage	29.0%
Flash usage	41.2%

2 Block Diagram



3 Component List

Instance ID	Catalog Name	Iface	GPIO Summary
soil_1	Capacitive Soil Moisture Sensor v1.2	ADC	AOUT=34
ds18b20_1	DS18B20	1-Wire	DQ=4
bme280_1	BME280	I2C 0x76	SDA=21, SCL=22
relay_1	Relay Module (single channel, 5V coil)	GPIO	IN=26
ili9341_1	ILI9341 2.8" TFT (SPI)	SPI	MOSI=23, MISO=19, SCK=18, CS=27, DC=16,

4 Wiring Table

Instance	Pin	GPIO	GPIO Notes
soil_1	AOUT	34	Input-only. No internal pull-up or pull-down. ADC1 channel 6.
ds18b20_1	DQ	4	ADC2 pin, used as digital I/O (WiFi restriction applies only to analog reads)
bme280_1	SDA	21	Default I2C SDA.
bme280_1	SCL	22	Default I2C SCL.

Instance	Pin	GPIO	GPIO Notes
relay_1	IN	26	ADC2 pin, used as digital I/O (WiFi restriction applies only to analog reads)
ili9341_1	MOSI	23	Default VSPI MOSI.
ili9341_1	MISO	19	Default VSPI MISO.
ili9341_1	SCK	18	Default VSPI SCK.
ili9341_1	CS	27	ADC2 pin, used as digital I/O (WiFi restriction applies only to analog reads)
ili9341_1	DC	16	Default UART2 RX. Good general-purpose I/O.
ili9341_1	RESET	17	Default UART2 TX. Good general-purpose I/O.

5 Pin Map

All ESP32 GPIO assignments at a glance:

GPIO	Assigned To	Pin / Function
4	ds18b20_1	DQ
16	ili9341_1	DC
17	ili9341_1	RESET
18	ili9341_1	SCK
19	ili9341_1	MISO
21	bme280_1	SDA
22	bme280_1	SCL
23	ili9341_1	MOSI
26	relay_1	IN
27	ili9341_1	CS
34	soil_1	AOUT

6 Bus Configuration

6.1 I2C (i2c0)

Parameter	Value
SDA	GPIO 21
SCL	GPIO 22
Frequency	400 kHz
bme280_1 address	0x76 (118)

6.2 SPI (spi0)

Parameter	Value
MOSI	GPIO 23
MISO	GPIO 19
SCK	GPIO 18
Frequency	40 MHz
CS (ili9341_1)	GPIO 27

7 Component Notes

7.1 soil_1 — Capacitive Soil Moisture Sensor v1.2

Must use an ADC1 pin (GPIO32–36, 39) when WiFi is enabled — ADC2 pins are disabled by the WiFi driver. Output does NOT reach 3.3V: typical range is ~2.5–2.9V in dry air and ~1.0–1.5V submerged in water at 3.3V supply. Always calibrate — thresholds vary by sensor unit and PCB revision. Some clones use an NE555 oscillator that requires a 5V supply to function correctly; if the sensor output looks wrong at 3.3V, try powering VCC from 5V.

7.2 ds18b20_1 — DS18B20

Requires a 4.7k Ω pull-up resistor between DQ and VCC — the ESP32 internal pull-up is too weak for reliable 1-Wire communication. The DallasTemperature library wraps OneWire for convenience; both must be in lib_deps. Temperature conversion at 12-bit resolution takes up to 750 ms — do not use delay(); use non-blocking requestTemperatures() + isConversionComplete().

7.3 bme280_1 — BME280

3.3V device — do not connect to 5V supply. SDO to GND selects address 0x76 (default); SDO to VCC selects 0x77. Always pass the address explicitly to begin() — the Adafruit library defaults to 0x77 but generic modules typically use 0x76. Many cheap modules are actually BMP280 (no humidity sensor): verify by checking the chip ID returned by begin(). The breakout board usually includes pull-ups on SDA/SCL; a second device on the bus does not need more pull-ups.

7.4 relay_1 — Relay Module (single channel, 5V coil)

The coil requires 5V — power from the DevKit's 5V (VIN) pin. Most modules are ACTIVE LOW: writing LOW energizes the relay, HIGH releases it. 3.3V compatibility warning: 5V active-LOW opto modules may not switch OFF reliably when the ESP32 drives the IN pin HIGH at 3.3V. Safe options: (1) remove the JD-VCC jumper and power VCC=3.3V / JD-VCC=5V separately so the opto input is 3.3V-referenced; (2) use a relay module rated for 3.3V coil; (3) add a transistor driver (e.g. 2N2222 + 1k Ω base resistor). Firmware tip: write HIGH to the IN pin before calling pinMode(OUTPUT) to prevent the relay from clicking at boot when the GPIO floats low during initialization.

7.5 ili9341_1 — ILI9341 2.8" TFT (SPI)

Configure TFT_eSPI exclusively through build_flags in platformio.ini — never edit files inside .pio/libdeps/. Required build flags: -D USER_SETUP_LOADED -D ILI9341_DRIVER -D TFT_MOSI=23 -D TFT_SCLK=18 -D TFT_CS=<pin> -D TFT_DC=<pin> -D TFT_RST=<pin> -D SPI_FREQUENCY=40000000. Use LVGL v9 API only (NOT v8). Use a partial draw buffer (~1/10 screen = 320*24*2 = ~15 KB).

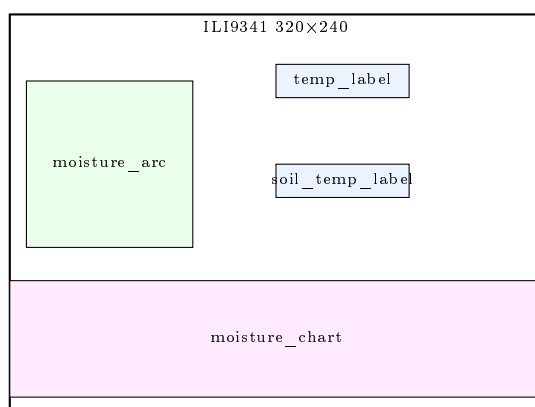
8 LVGL Dashboard Layout

Driver: ili9341 Display instance: ili9341_1

Firmware built with **LVGL v9.2.2** (TFT_eSPI back-end). RAM usage: **29.0%**, Flash usage: **41.2%**.

Widget ID	Type	Binding	X	Y	W	H	Min	Max
moisture_arc	arc	soil_1.moisture	10	40	100	100	0	100
temp_label	value	bme280_1.temperature	160	20	—	—	—	—
soil_temp_label	value	ds18b20_1.temperature	160	80	—	—	—	—
moisture_chart	chart	soil_1.moisture	0	160	320	70	0	100

8.1 Dashboard Canvas (320×240)



9 Firmware Build Result

Item	Value
LVGL version	9.2.2
TFT_eSPI version	2.5.43
Build status	SUCCESS
RAM usage	29.0%
Flash usage	41.2%